Terrain Classification for Small-Legged Robots Using Deep Learning on Tactile Data

Team F2 | Two-Page Experimental Project Report

1. Problem Statement

Robots moving over varied terrain need to recognize the surface beneath them to support reliable locomotion. This project evaluates tactile-data-based classification of eight terrain categories: asphalt, cork, grass, gravel, lab, laminate wood, pebble and sand. Classical machine-learning models are compared with a convolutional neural network (CNN).

2. Dataset and Preprocessing

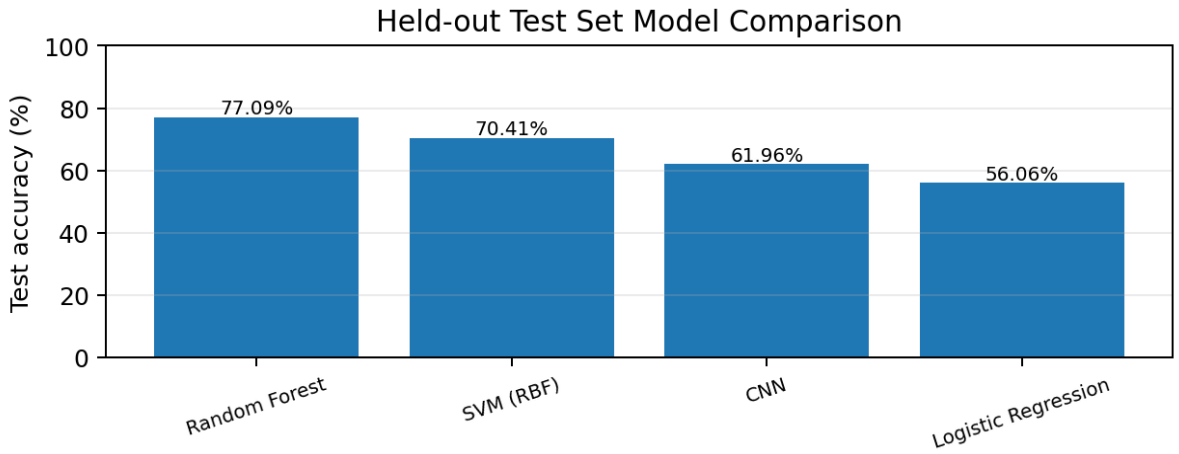
The dataset contains 5,990 samples and 447 input features per sample: 414 tactile readings arranged as 6 channels × 69 readings, plus 33 engineered features. A stratified split was used: 4,193 training samples (70%), 898 validation samples (15%) and 899 test samples (15%). Normalization parameters were fitted on training data only and then applied to validation and test data.

3. Models and Evaluation

The evaluated models were Logistic Regression, an RBF-kernel Support Vector Machine (SVM), Random Forest and a CNN. The first three used all 447 features; the CNN processed the tactile readings with convolutional layers and concatenated the learned representation with all 33 engineered features. Models were compared using accuracy and macro F1-score. Validation data was used during model selection; the held-out test set was used for final evaluation.

4. Final Test Results

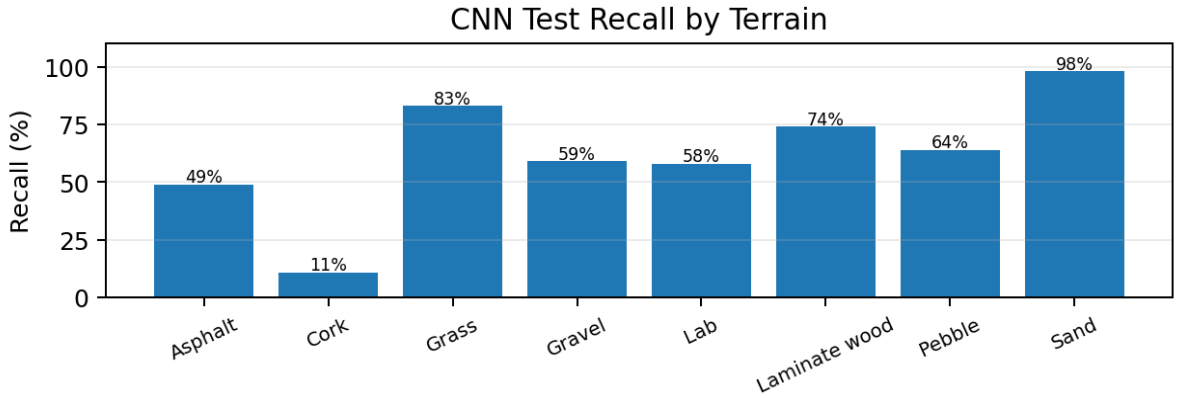
Model	Test accuracy	Macro F1
Random Forest	77.09%	0.7698
SVM (RBF)	70.41%	0.7093
CNN	61.96%	0.5900
Logistic Regression	56.06%	0.5593



Random Forest achieved the strongest performance in this experiment, with 77.09% test accuracy and a macro F1-score of 0.7698.

5. CNN Performance and Error Analysis

The CNN achieved 61.96% test accuracy and a macro F1-score of 0.5900. Per-class recall was strongest for sand (98%) and grass (83%), while cork recall was only 11%. Recall for asphalt, gravel, lab, laminate wood and pebble was 49%, 59%, 58%, 74% and 64%, respectively. These results show that overall accuracy hides differences in how well individual terrain classes are recognized.



6. Discussion

Random Forest outperformed the CNN by approximately 15.13 percentage points in test accuracy. One possible interpretation is that the tabular engineered features and their nonlinear interactions are effectively handled by tree-based learning. This experiment does not show that Random Forest is universally superior to CNNs; the conclusion is limited to this dataset, split, preprocessing and model configuration.

7. Limitations and Future Work

Results are based on one stratified split and may vary with different data splits or deployment conditions. The CNN's low cork recall is a key limitation. Future work could test repeated stratified splits or cross-validation, inspect feature importance, evaluate alternative architectures, and investigate class-specific errors. Real-world testing is needed to assess robustness across robots, sensors and environments.

8. Conclusion

This project compared three classical classifiers and a CNN for eight-class terrain classification using tactile data. Random Forest performed best among the evaluated models, reaching 77.09% held-out test accuracy and 0.7698 macro F1. The CNN reached 61.96% accuracy, with strong recall on sand and grass but weak recall on cork. The results support benchmarking deep learning against classical methods rather than assuming a neural network will perform best.

Reproducibility: The accompanying notebook contains the preprocessing, model training and evaluation workflow. Results in this report are based on the experiment outputs provided for this project.